

Ishan Yateen Joshi

AI & Machine Learning Engineer

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EDUCATION

MIT World Peace University

Bachelor of Technology in Computer Science and Engineering

- CGPA: 7.98/10.0
- Relevant Coursework: AI & Machine Learning, Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Linear Algebra, Statistics

Pune, Maharashtra

Expected May 2026

EXPERIENCE

Senior Computer Vision Intern

Jan 2026 – Present

Thelios AI – NBA Partnership

Pune, India

- Building a **scalable video-intelligence platform** on self-supervised transformer models, producing reusable multi-level representations for fast downstream analytics in sports and video understanding.
- Designing a **dual-path self-supervised video model** (Masked Autoencoders + Joint Embedding Predictive Architecture) to learn spatiotemporal representations from unlabeled broadcast footage; explored encoder designs and masking strategies to mitigate representation collapse.
- Built end-to-end PyTorch training pipelines for video transformers with patch tokenization, mixed-precision (AMP), and multi-GPU distributed training, tuned to fixed compute and memory budgets for **production-ready, repeatable experimentation**.

Computer Vision Intern

Jul 2025 – Jan 2026

Thelios AI – NBA Partnership

Pune, India

- Developed **real-time multi-object detection and tracking** pipelines for NBA broadcast footage using YOLOv11 variants, segmentation (SAM2), and pose-keypoint extraction for players and court elements.
- Built an **object-tracking evaluation framework** (identity consistency, ID-switch rates over long sequences); profiled GPU utilization and speed-accuracy tradeoffs to resolve end-to-end latency bottlenecks.
- Supported large-scale **data preparation and annotation** and contributed to event-detection tasks; built internal tools including the frontend for a **RAG-based research-paper assistant**.

SOC Intern

Jul 2025

Microscan Communications Pvt Ltd

Pune, India

- Monitored security alerts with SIEM tools, investigated suspicious activity via log, endpoint, and network-traffic analysis, and supported incident response and VAPT efforts following defined playbooks.

PROJECTS

Self-Supervised Video Representation Learning (Dual-Path V-JEPA + MAE) | PyTorch, Video Transformers 2026

- Prototyped a two-branch self-supervised learner: a reconstruction branch predicting masked spatiotemporal patches and a predictive branch regressing target embeddings from context embeddings in latent space.
- Studied collapse-avoidance strategies (EMA target encoder, stop-gradient, predictor bottleneck) and ablated masking ratios, block sizes, and frame-sampling strides toward a domain foundation backbone.

AI-Based Anomaly Detection for Network Intrusion | Python, TensorFlow, Scikit-learn

Feb – Apr 2025

- Built ensemble classifiers (Random Forest + GBM) with automated class-imbalance handling and hyperparameter search; achieved **94.44% accuracy on KDD Cup 99** and **88.39% on NSL-KDD**, with a real-time detection loop tuned for low false-positive rate.

Brain Tumor Detection from MRI | PyTorch, TensorFlow, CNNs, Flask

Jan 2024

- Built a CNN pipeline for automated tumor classification reaching **89.4% accuracy** with preprocessing and augmentation for limited, class-imbalanced data; deployed a real-time Flask inference app returning predictions with confidence scores.

Bird Migration Analysis on BNHS Data | Python, Data Science, Statistical Modeling

Jul 2024 – Apr 2025

- Modeled climate and geography as drivers of route shifts across large-scale Bombay Natural History Society migration records, supporting biodiversity-conservation research.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, Bash

Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, Keras, NumPy, Pandas

Computer Vision: OpenCV, YOLOv8/v11, CNNs, ViT/ViViT, object detection, multi-object tracking, segmentation (SAM2), pose estimation, real-time video pipelines

ML & Systems: self-supervised learning (MAE, JEPA/V-JEPA), CUDA/GPU & mixed-precision (AMP) training, multi-GPU distributed training, latency profiling, inference serving, RAG / agentic AI, FFmpeg

Tooling: Linux, Docker, Git, REST APIs, Jupyter, Colab; Matplotlib, Seaborn, Plotly, Streamlit, Tableau; MySQL, PostgreSQL, MongoDB

Certifications: Google Cybersecurity Certificate, CS50x, Claude Code 101 | **Spoken:** English, Hindi, Marathi (Fluent); German (Elementary)